

Replace " $c = a ** b$ " to " $c = \text{np.power}(a, b)$ "

① User code :

① $c = a ** b$



② Use LibCST turn into CST



③ add more energy info to it



④ turn each node of CST to
ML input feature



⑤ put into trained ML model
get replacement patch code



⑥ Use Libcst transformer class
to modify the resource code in
CST

nodes:
targets = [AssignTarget("c")]
Value = BinaryOperation(
left = "a"
operation = Pow()
right = "b"
③ energy_cost: xxx



④ converted ML input

ML model

⑤

ML output

$c = \text{np.power}(a, b)$

⑥